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ORIGINAL ARTICLE

A digital twin-based layout optimization method for discrete manufacturing workshop

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Abstract

In the discrete manufacturing workshop, an unreasonable workshop layout has a significant impact on the production efficiency, which can result in a large distance between operations, low utilization of tooling, and the backlog of products in production process. However, the existing optimization algorithms for workshop layout rarely take into account the real-time feedback of physical information from the workshop such that the layout cannot be self-adjusted to realize the optimum during manufacturing. Thus, this paper focuses on a discrete manufacturing workshop layout optimization based on digital twin, in which the workshop layout problem is solved by twin data fusion, information and physical interaction fusion, and data analysis and optimization. First, a sub-framework of digital twin-based workshop partitioning is established and the workshop partitioning is optimized via simulation analysis. Second, a sub-framework of digital twin-based equipment layout optimization is presented, in which equipment layout decisions are made by real-time data collection and value-added processing of twin data. Then, a sub-framework of digital twin-based distribution route optimization is developed for the workshop. Finally, the proposed method is applied to a welding production workshop and increased the production capacity o by 29.4%.

Keywords Discrete manufacturing workshop · Workshop layout optimization · Digital twin · Welding production

1 Introduction

With the rapid development of "Industry 4.0," all countries' attention is attracted by "smart manufacturing" the realization of true intelligent manufacturing [1, 2]. One of the most important parts of intelligent manufacturing is discrete manufacturing workshop. Generally, it contains many changes and uncertain factors because of its multiple sub-processes. And the layout optimization of discrete manufacturing workshop can promote the realization of intelligent manufacturing.

The discrete manufacturing workshop layout plays a decisive role in manufacturing logistics. A good layout optimization can reduce the cost of the manufacturing plant by 10–30% [3, 4]. Many discrete manufacturing workshops have the prominent problem of workshop layout, which is a problem that researchers have been trying to solve. In manufacturing,

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layout design refers to the most appropriate arrangement of manufacturing equipment in a defined area. When the layout design is unreasonable, it may lead to problems such as overlong distance between working stations, underutilization of tooling, and backlog of WIP (work in process), which will seriously affect the production efficiency.

In recent decades, the optimization model of manufacturing system layout has experienced a rapid development because of the development of operations research and computing science. A large number of optimization models have been built [5]. Diefenbach et al. [6] developed an accurate algorithm that is suitable for two targets and optimized the workshop layout of the U-shaped order picking area. Lu et al. [7] build an alternative model for the traditional layout optimization algorithm of the storage pool. Liao et al. [8] optimized a workshop layout by using system layout design (SLP) method. Liu et al. [9] optimized an assembly workshop layout through computer-aided design and optimization algorithms. Li et al. [10] solved the cache configuration and unreasonable equipment layout of a production workshop by improved NSGA-II algorithm. Scholars also adopted the SLP method to optimize the subway construction layout [11].

However, most researchers use exact algorithms or heuristics to optimize discrete workshop layouts, which exist several limitations. First, during the production process, due to equipment failure or WIP backlog changes at any time, some algorithms are no longer applicable and have not traversed all possible results. Second, the results of traditional optimization are often not the same as the global optimal solution due to the lack of real-time feedback of physical information. Third, the optimal optimization effect cannot be obtained. Thus far, there are few related papers that consider the digital workshop layout of intelligent manufacturing [12]. In order to enhance the real-time feedback capability, optimization of the discrete manufacturing workshop layout needs to be combined with the advanced technologies. The key technologies include IOT (internet of things), CPS (cyber-physical system), big data analysis, and so on [13, 14].

Digital twin makes full use of data such as physical models, sensor updates, and operation history. It integrates multi-disciplinary, multi-physical quantities, multi-scale, and multi-probability simulation processes [15, 16]. Essentially, it is a functional system that combines physical entities and twin models and can be continuously optimized [17]. Therefore, digital twin can be employed to deal with some crucial problems in manufacturing workshops. For example, Tao et al. [18, 19] proposed the conceptual model of the digital twin workshop and analyzed its implementation method. At the same time, the interaction of physical workshop and virtual workshop based on digital twin is discussed and summarized [20]. And it provides a reference for the implementation of digital twin technology in intelligent manufacturing workshops. At the practical level, the Reutlingen Business School in Germany established a prototype of a digital twin factory based on cloud and app applications [21]. Bao and others in China proposed a digital twin-based workshop modeling framework [22]. Nevertheless, there are a few studies on applying digital twin to the layout optimization of discrete manufacturing workshops. Therefore, this paper puts forward a new workshop layout optimization method based on digital twin to improve the overall quality and efficiency of the discrete manufacturing process.

For discrete manufacturing workshops, the layout problems that lead to low production capacity mainly include the following aspects: (1) the confusion of logistics, the extension of logistics path, and the increase of production cycle due to unclear partitioning of workshop and (2) the overlong distance between work stations and the inefficient production process due to the unreasonable distribution of equipment and facilities; (3) The accumulation of WIP and the low utilization rate of tooling due to the unreasonable material distribution route and the low accuracy of material distribution. In order to solve the above workshop layout problems, a digital twin-based discrete manufacturing workshop layout optimization method is established based on three sub-frameworks of workshop partitioning, equipment layout optimization, and distribution route optimization. Then, the proposed method is applied to a welding production workshop.

The remainder of this paper is organized as follows. Section 2 explains the concept of digital twin and provides a detailed description of main components of digital twin system. Section 3 proposes a digital twin-based layout optimization method for discrete manufacturing workshop and illustrates the frameworks. Section 4 applies a welding production workshop to test the proposed approach. Finally, section 5 presents conclusions and future work.

2 Digital twin

Digital twin is an integrated system that can simulate, monitor, calculate, regulate, and control the system status and process [23]. It has the characteristics of individualization, high efficiency, and highly quasi-real. In addition, digital twin is developed by data acquisition, virtual manufacturing technologies, based on the control, computation, and communication units. There are three main components in digital twin, i.e., physical layer, digital layer, and information fusion layer that connect two layers.

2.1 Physical layer

Physical layer refers to the physical entities of workshops such as people, machines, and objects and the workplace where the entity is located: the production activities and design platforms. Physical layer is always equipped with sensors and

other equipment to collect and transmit the data of the underlying equipment.

Compared with traditional workshop, the physical layer workshop has the ability to perceive and integrate real-time data, as well as the ability to integrate the human-machine-physical-environment elements of the workshop. The physical layer encapsulates multi-type, multi-scale, and multi-granular physical workshop data. In addition, it realizes the unified and standardized processing of operable and traceable data. And it realizes the fusion of multi-source and multi-modal data in the physical workshop through operations such as data classification, association, and combination. In addition, the production factors of the physical layer need to be integrated to adapt to the complex and changing environment [24]. Compared with the traditional human-centered decision-making workshop, the "human-machine-physical-environment" factor-integrated physical layer has greater flexibility, adaptability, robustness, and intelligence [25].

2.2 Digital layer

Digital layer refers to the digital twin model and its corresponding virtual layout optimization activities [26]. It includes the twin data storage, analysis, calculation of production activities in the virtual layer, and the prediction, comparison, and decision-making in optimization.

The digital layer includes physical models of production factors such as human, machine, material, and environment [27]. It is based on the rules of evaluation, optimization, prediction, and source tracing established by various operation and evolution laws. Before production, the digital layer simulates the entire production process based on an iterative simulation analysis of the production plan. In this way, possible problems in the production plan can be discovered in time, and real-time adjustments and optimizations can be made. During production, the digital layer continuously accumulates real-time data and knowledge of the physical layer [28]. Under the premise of high fidelity of the physical layer, the operation process is continuously adjusted and optimized. The related information in the digital layer can be superimposed and interacted with the physical layer in real time to realize the fusion, integration, and real-time interaction between the digital layer and the physical layer.

2.3 Information fusion layer

Information fusion layer refers to the optimization through cyber-physical interactive fusion and analysis of twin data. In this paper, it helps to solve the problem of large interprocedure distance, low efficiency of tooling, and backlog of work-in-progress caused by unreasonable layout of the workshop. It also provides overall data blending services in workshop partitioning, equipment layout optimization, and distribution route optimization.

Driven by data from the physical and digital layers, the information fusion layer generates a resource allocation plan and initial production plan that meets the task requirements and constraints. Before starting production, it revises and optimizes the production plan based on simulation, evaluation, and optimization data of the production plan from the digital layer. In the production process, the simulation, verification, and optimization results of the production status in the physical layer and the digital layer are continuously fed back to the fusion layer. Then, the fusion layer adjusts the production plan in real time to adapt to changes in actual production needs.

The fusion layer is the driver that realizes the operation of the physical layer and the digital layer and the interaction between the two [29]. On the basis of integration, the workshop twin data undergoes in-depth data fusion and constantly updates and expands its own data.

3 A digital twin-based layout optimization method for discrete manufacturing workshop

In order to solve the problem of unreasonable workshop layout, a digital twin-based layout optimization method is established for the discrete manufacturing workshop. And the sub-frameworks of workshop partitioning, equipment layout, and distribution route optimization are developed respectively based on digital twin.

3.1 Sub-framework of digital twin-based workshop partitioning

The workshop partitioning is a layout design method based on the predefined hardware equipment group of workshops. In workshop partitioning, workshop can be distributed by process or unit. Such as assembly technology group and polishing technology group distributed by process and complete equipment group distributed by units. Or other way of partitioning can be chosen according to the production needs and mutual the constraint relationship between modules. It helps solve the problem of large distance between processes caused by irrational workshop layout.

As shown in Fig. 1, the sub-framework of digital twin-based workshop partitioning is composed of physical layer, digital layer, and information fusion layer. (1) The physical layer mainly refers to the workshop entities such as humans, machines, objects, and production activities implemented by the entities, which is the physical mapping of the digital layer. (2) The digital layer mainly refers to the twin data and its processing, driving data, digital twin models, virtual layout

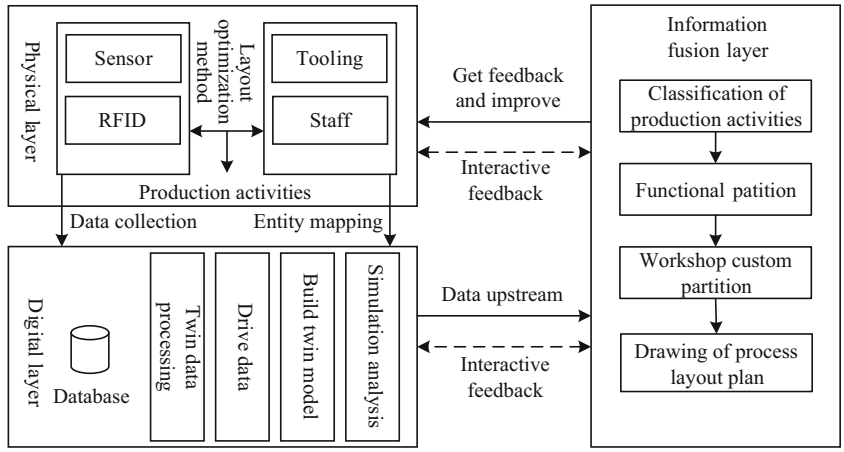


Fig. 1 Sub-framework of digital twin-based workshop partitioning

optimization data, and other databases. (3) The information fusion layer refers to the functional implementation process of production activity classification, functional partitioning, workshop custom partition, and drawing of process layout plans. The sub-framework of digital twin-based workshop partitioning operates as follows: physical entities are adjusted according to the layout optimization method, the data collected from the underlying equipment of physical layer is transmitted to the digital layer through the processing of information fusion layer, through the simulation analysis, the digital layer and the simulation results are fed back to the physical layer simultaneously through information fusion layer, and the information fusion layer continues to interact with the physical layer and the digital layer until the problem of unreasonable workshop division is effectively solved [30].

Layout problem such as large distance between processes, low efficiency of tooling, and backlog of work in progress exists in many discrete manufacturing workshops, which seriously affect the production efficiency. Reasonable layout design can effectively reduce material handling costs and time and reduce product production costs. The cost of material handling generally includes the cost of material handling inside and outside the unit. The expression is as follows [31]:

$$\min Z_{1n} = C_{inter} + C_{intra} \tag{1}$$

$$C_{inter} = \sum_{m_i=1}^{M}\sum_{m_j=1}^{M}\sum_{l=1}^{L}\sum_{l'=1}^{L}\sum_{p=1}^{P} C_{m_i m_j} d_{ll'} D_p X_{m_i l} X_{m_j l'} \tag{2}$$

$$C_{intra} = \sum_{m_i=1}^{M}\sum_{m_j=1}^{M}\sum_{c=1}^{C}\sum_{c'=1}^{C}\sum_{p=1}^{P} C_{cc'} d_{cc'} D_p X_{m_i c} X_{m_j c'} \tag{3}$$

Among them: m_i, m_j are the device numbers, $\{m_i, m_j = 1, 2, \ldots, M\}$; l、l' are the position numbers within the unit, $\{l, l' = 1, 2, \ldots, L\}$; c, c' are manufactured units, $\{c, c' = 1, 2, \ldots, C\}$; $d_{ll'}$ is the distance between position l and position l' in the unit; $d_{cc'}$ is the distance between unit c and c'; $C_{m_i m_j}$ is the cost per unit of movement between facility m_i and facility m_j in the unit; $C_{cc'}$ is the total cost per unit of movement between unit c and c'; and D_p is the demand for the pth product.

$$X_{m_i l} = \begin{cases} 1, \text{if the facility } m_i \text{ is placed in position l} \\ 0, \text{else} \end{cases} \tag{4}$$

$$X_{m_i c} = \begin{cases} 1, \text{if the facility } m_i \text{ is placed in unit c} \\ 0, \text{else} \end{cases} \tag{5}$$

By re-planning the layout of the workshop, adjusting the equipment and facilities at the production site, and moving small work stations that are far away from work to the main line of the shift team, we reduce the distance between workshop processes. It can reduce the distance $d_{cc'}$ between the unit c and unit c', thereby reducing the cost per unit of movement $C_{m_i m_j}$ between facility m_i and facility m_j in the unit and reducing the total cost per unit of movement $C_{cc'}$ between unit c and c'.

$$X_{m_j c'} = \begin{cases} 1, \text{if the facility } m_j \text{ is placed in position c}' \\ 0, \text{else} \end{cases} \tag{6}$$

After the workshop partitioning according to the production activities, the distance $d_{cc'}$ between the unit c and unit c' is shortened, and the total cost per unit of movement $C_{cc'}$ between the unit c and c' is reduced, thereby reducing the material moving cost C_{intra} outside the unit.

3.2 Sub-framework of digital twin-based equipment layout optimization

The configuration and combination of machines, equipment, transportation channels, buildings, sites, etc. are optimized by the sub-framework of digital twin-based equipment layout optimization according to production need, workers demand, logistics, and information flow. The sub-framework of digital twin-based equipment layout optimization consists of physical entities, digital models, twin data, and layout optimization decisions, as shown in Fig. 2.

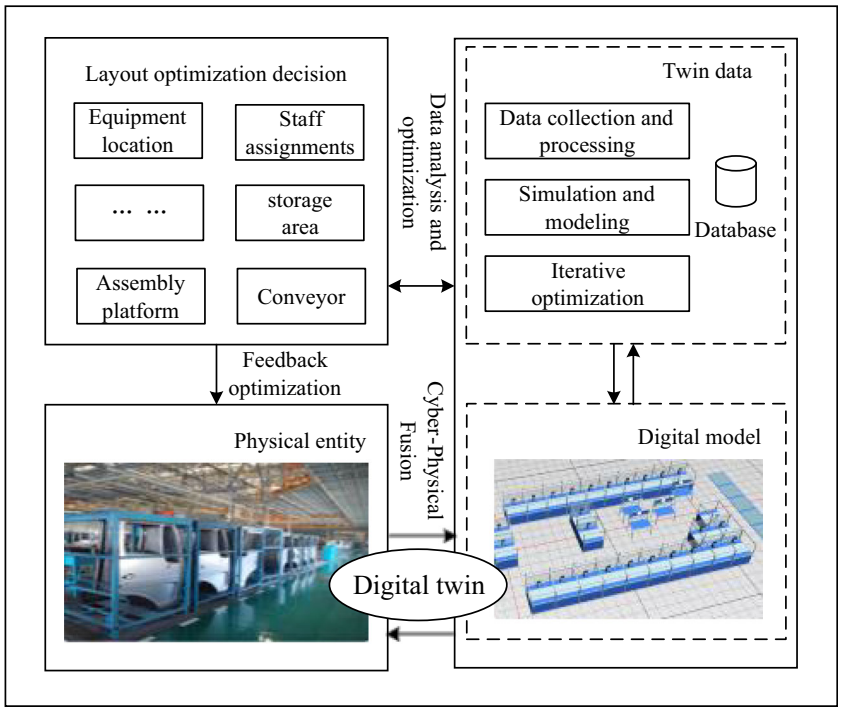


Fig. 2 Sub-framework of digital twin-based equipment layout optimization

The differences between proposed optimization and traditional equipment layout optimization are as follows: (1) physical entities are mapped to digital models through information integration, model construction, two-dimensional monitoring, and three-dimensional monitoring. The physical workshop and virtual workshop run simultaneously [32]. (2) Real-time data collection, twin data value-added processing, and workshop status monitoring are carried out based on data collection technology. (3) Considering equipment selection, personnel allocation, storage area, etc., the data is continuously and iteratively optimized to obtain optimal layout optimization decisions, so as to further solve the problems of overlong distance between work stations and unreasonable distribution of equipment and facilities.

Take the partitioning as a unit, the distance $d_{ll'}$ between the position l and the position l' in the unit, the cost per unit of movement $C_{m_i m_j}$ between facility m_i and facility m_j in the unit, and the demand D_p for the pth product; the cost C_{inter} of material moving outside the unit satisfies the expression [31].

$$X_{m_j l'} = \begin{cases} 1, \text{if the facility } m_j \text{ is placed in position } l' \\ 0, \text{else} \end{cases} \quad (7)$$

After the equipment layout is optimized, the distance $d_{ll'}$ between the position l and the position l' in the unit is shortened, and the cost per unit of movement $C_{m_i m_j}$ between facility m_i and facility m_j in the unit is reduced, thereby reducing the material moving cost C_{inter} inside the unit.

3.3 Sub-framework of digital twin-based distribution route optimization

The optimization of distribution routes includes the following aspects: (1) reduce the time required for distribution and save distribution mileage, (2) improve the overall logistics speed, and deliver WIP to the next equipment link accurately and efficiently. (3) Make distribution arrangements reasonable, help workshops reduce costs, improve overall competitiveness and create better benefits [33].

As shown in Fig. 3, sub-framework of digital twin-based distribution route optimization is established. It is formed by the interaction between physical and virtual distribution routes. The physical distribution route and the virtual distribution route are mapped to each other to realize simulation modeling and collaborative processing, so as to achieve workshop distribution route optimization and further solve the problems of low tooling use efficiency and the WIP backlog [34].

There are a total of K distribution routes in the optimization plan, W is an index that reflects the workload of all distribution routes in the workshop in the optimization plan, ρ_k is the workload weight of the kth distribution route, S_k is the delivery mileage (meters), T_k is the delivery volume (piece), and U_k is the number of work stations (pieces). The value of W in the optimization scheme satisfies the following expression:

$$\min W = \sum_{k=1}^{K} (\rho_k \cdot S_k \cdot T_k \cdot U_k) \quad (9)$$

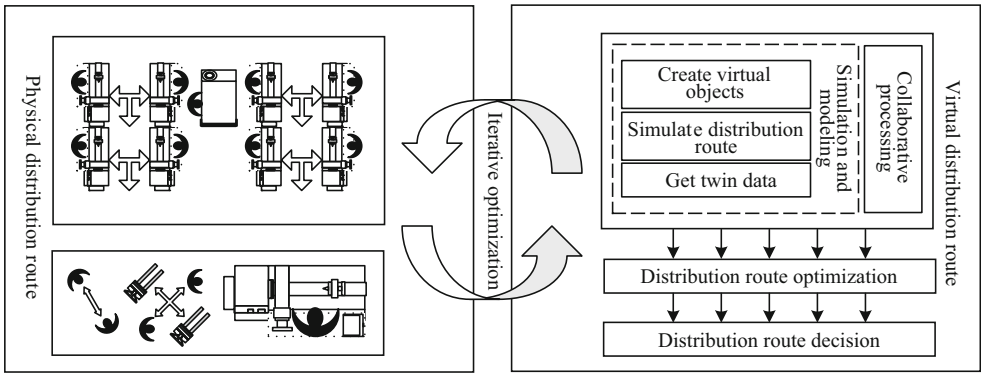


Fig. 3 Sub-framework of digital twin-based distribution route optimization

3.4 Digital twin-based discrete manufacturing workshop layout optimization method

Combining layout optimization methods and simulation technologies, we propose a digital twin-based discrete manufacturing workshop layout optimization method based on the above three sub-frameworks: workshop partitioning, equipment layout optimization, and distribution route optimization, as shown in Fig. 4, so as to provide an overall solution for workshop layout optimization problems such as unreasonable layout of equipment and facilities, low tooling efficiency, and WIP backlog.

The digital twin-based discrete manufacturing workshop layout optimization method mainly includes (1) physical layer, refers to the machines, workers, sensors, raw materials, workpieces, products, and other physical entity as well as the workshop production activities and workshop itself. The physical layer transmits the on-site real-time data to the digital layer through the information fusion layer. (2) Digital layer refers to the digital twin model and its corresponding virtual layout optimization activities, including the twin data storage, analysis, and calculation of production activities in the virtual environment, and the prediction, comparison, and optimization decision of the digital model. The digital layer transmits the twin data obtained from simulation to the physical layer through the information fusion layer. (3) Information fusion layer refers to twin data fusion, information physical interaction fusion, data analysis, and optimization decision. Information fusion layer is responsible for providing overall data fusion service for workshop partitioning, equipment layout optimization, and distribution route optimization.

4 Case study

In this paper, a typical discrete manufacturing workshop—welding production workshop is studied as an optimization case. By situation analysis, optimization plan making, virtual simulation, optimization plan implementation, and result analysis, the proposed layout optimization method is applied to a welding production workshop of a manufacturing company.

4.1 Partitioning of welding production workshop based on digital twin

According to the sub-framework of digital twin-based workshop partitioning, the workshop partitioning plan of welding production workshop is made based on the layout optimization method. The distribution of entities in physical layer is adjusted based on partitioning plan. Through information fusion layer, the on-site data is transmitted to the digital layer to established simulation modal to analyze the rationality of the partitioning of physical layer. Then, the results from simulation analysis are fed back to the physical layer through information fusion layer to improve the workshop partitioning plan [35].

After the continual improvement to the physical layer partitioning, the welding production workshop is divided into work area, storage area, and rest area. In the optimized workshop, each kinetic energy pipeline is shortened, the equipment and tooling that have not been used for a long time are removed, and safety protection measures are taken.

4.2 Equipment layout optimization of welding production workshop based on digital twin

According to the sub-framework of digital twin-based equipment layout optimization, the factors such as equipment selection, personnel allocation, and storage location are analyzed. The product, product flow, production plan, process improvement, and equipment improvement are studied. The equipment layout optimization plan is presented according to the layout optimization decisions. The virtual workshop is established according to the equipment layout optimization plan. The twin data from simulation result of virtual workshop is mapped back to the physical workshop to remake the equipment layout optimization plan. The above iterative processes

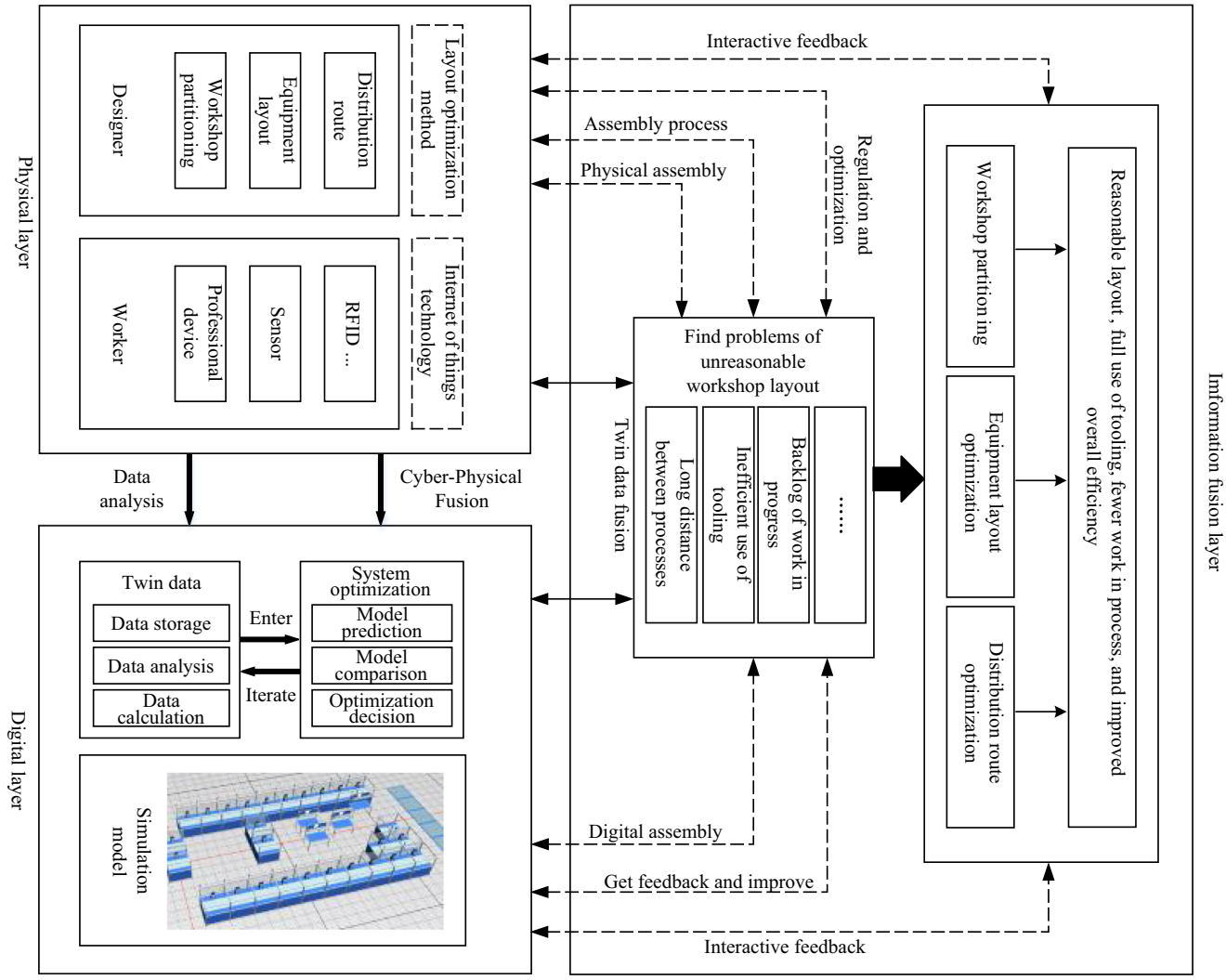


Fig. 4 Framework of digital twin-based discrete manufacturing workshop layout optimization method

are repeated until the desired equipment layout plan is achieved [36].

According to the new plan of the welding production workshop, the pilot teams are set up to re-plan the physical equipment layout. The equipment and facilities are arranged in parallel layout form. And the small work stations located far from main line are moved back to the line.

4.3 Distribution route optimization of welding production workshop based on digital twin

According to the sub-framework of digital twin-based distribution route optimization, the physical distribution routes are simulated, and it generates real-time workshop data. The virtual distribution routes are established, and it generates twin data to interact with the physical workshop. The iterative feedback between physical and virtual distribution routes makes the distance of the WIP distribution route as small as possible. Through the routing decision, the optimization plans are filtered. The purpose of getting the iterative optimization of the WIP distribution route of the welding production workshop is achieved [37].

Aiming at the problems of large inventory of WIP, excessive tooling, and clutter, distribution routes of the welding production workshop are optimized. And the number of its WIP and workstations is reduced. So, the problems of tooling waste and the WIP backlog are improved, the goals of multi-frequency, low-volume and highly accurate distribution are achieved.

4.4 Virtual model simulation of welding production workshop

A good workshop layout has a significant impact on the production efficiency. By applying the digital twin-based discrete manufacturing workshop layout optimization method, we can improve the product quality and production efficiency,

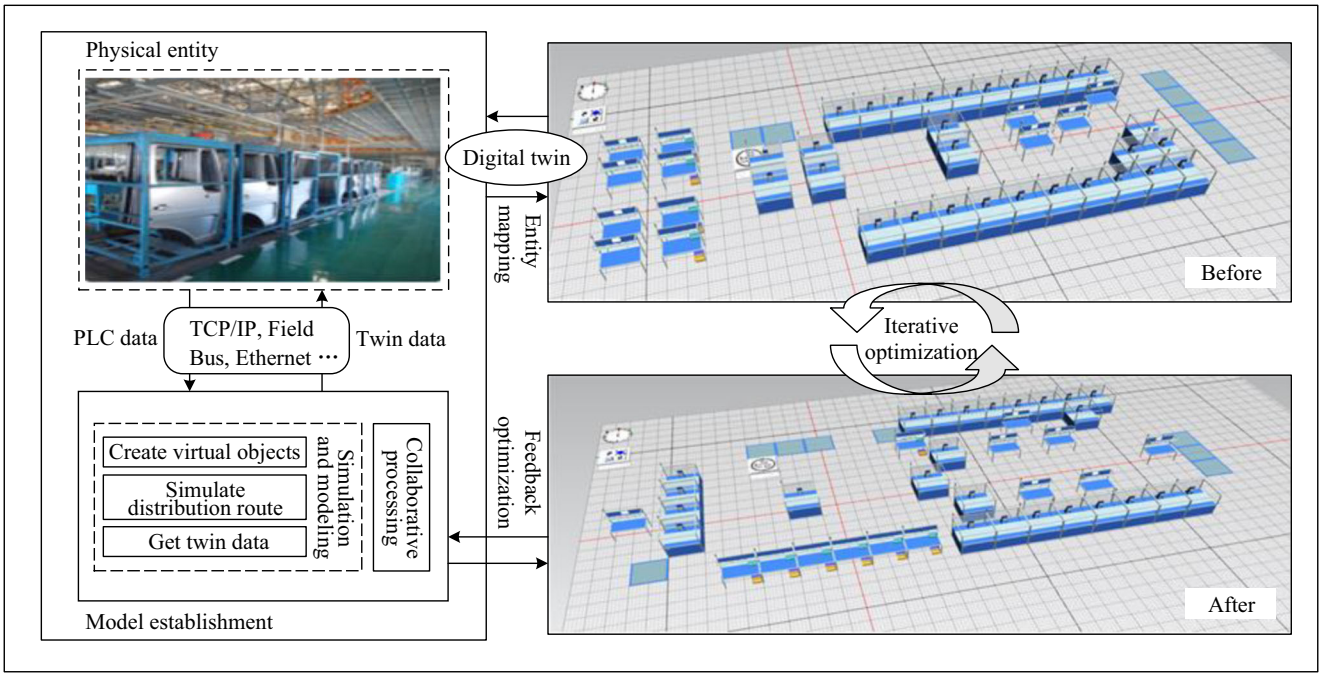


Fig. 5 Original-optimized model of workshop layout

improve the level of the on-site management and the operation safety, reduce the costs, and meet the market demands.

4.4.1 Establishment of virtual model

PLC control logic is prepared at the data acquisition terminal of physical workshop as the simulation input of virtual model. The data collected by the PLC includes layout of the workshop, process routes of the parts, and the resource equipment (machine tools, robots, logistics systems, etc.). Plant simulation is used to map these data to the virtual layer to establish the virtual model of physical workshop. The communication between the virtual model and the PLC is realized through TCP/IP, fieldbus, Ethernet, etc. According to the three subframeworks of the layout optimization method, the virtual model is optimized iteratively. And the optimal partitioning, optimal equipment layout, and the optimal distribution route are obtained after a certain time of optimization. Then, the final results will be transmitted back to the physical layer to rearrange the entities in physical workshop, as shown in Fig. 5.

4.4.2 Analysis of simulation results

By analyzing Fig. 6, it can be seen that the number of products produced in the same time after digital twin optimization is greater. And with the increase of time, the number of products produced in the same time increases faster. The interval between the completion of two adjacent products is more stable. From the analysis in Table 1, it can be seen that the layout optimization based on digital twin has a greater production capacity than the original workshop. The decrease in the number of WIP indicates that the tooling utilization has been improved after optimization. And the increase of production frequency indicates that the WIP is better controlled. It further

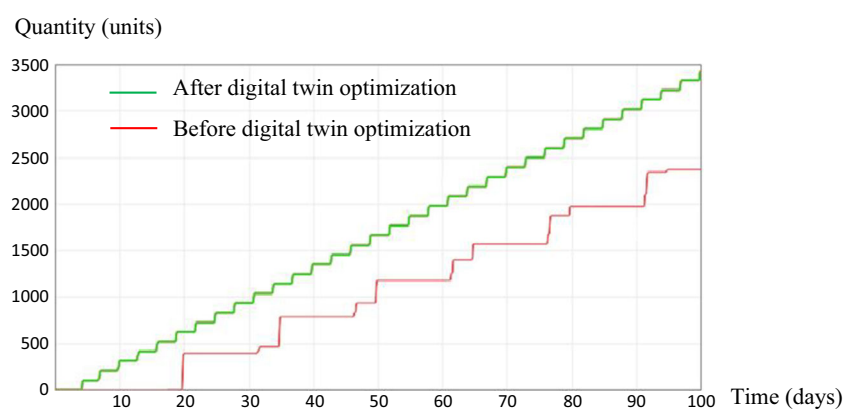


Fig. 6 Comparison of production capacity before and after digital twin optimization in the door line

Table 1 Comparison of the effects before and after digital twin optimization (at 50 days)

Workshop	Production capacity/set	Work in progress/set
Before digital twin optimization	1200	45
After digital twin optimization	1700	10
Effect comparison	Increased 29.4%	Down 77.8%

reflects the superiority of digital twin-based discrete manufacturing workshop layout optimization method.

4.5 Results analysis

The digital twin-based layout optimization method is applied to optimize the layout of the door line and the front wall line of a welding production workshop. The following results are obtained:

(1) Shorter distance between processes and less man-hour input

As shown in Fig. 7, the distance between the stations of the door line after the optimization is kept below 6 m. And the frequency with the distance of 3 m is the highest. Compared with the door line stations before optimization, the number of stations after optimization is reduced. Moreover, the average distance between stations is reduced, which is indicating that the digital twin-based layout optimization method shortened the distance between processes.

According to on-site records, the door line is planned to produce 120 doors per day, and there are 7 workers at the door line. Through the improvement to the on-site layout and process flow, the front wall line saved 67 steps of walking distance between processes (each step saves 0.5 s of working time). Sixty units of front wall are planned to be produced per day. There are 11 workers at the front wall. The cost is 2.3 yuan per hour per person, and the work time is 228 days per year. By shortening the distance between processes and reducing the transfer cycle, the total work time is reduced by 114.5 min per day. And the man-hour investment is reduced by 11,846 yuan per year.

(2) Fewer moving tools and fewer WIP

The cost of each tooling is 1,100 yuan per set, and the cost of post-maintenance of each tooling is 1,100 yuan per year. Through the independent improvement using excess materials, the number of mobile tooling and fixed tooling are reduced. The manufacturing cost of each material box is 70 yuan per piece. The workers independently made small material boxes with excess materials, which saved the original material boxes. The cost of each WIP backlog is 5,400 yuan. The number of WIP between processes is reduced, which decreased the occupation of WIP funds.

(3) Workshop layout optimization and workshop partitioning

After the implementation of the improvement, the layout of the workshop is optimized, and the working space is fully utilized. The kinetic energy pipelines are shortened, and safety protection measures are taken. Site cleanliness and site safety factors have been improved. The comparison of the process layout of the door line before and after optimization is shown in Fig. 8. The optimization plan implementation effect shows that the digital twin-based discrete manufacturing workshop layout optimization method has a good effect in the welding production workshop.

By sorting out the optimization results after using the digital twin-based discrete manufacturing workshop layout optimization method, Table 2 is obtained as follows.

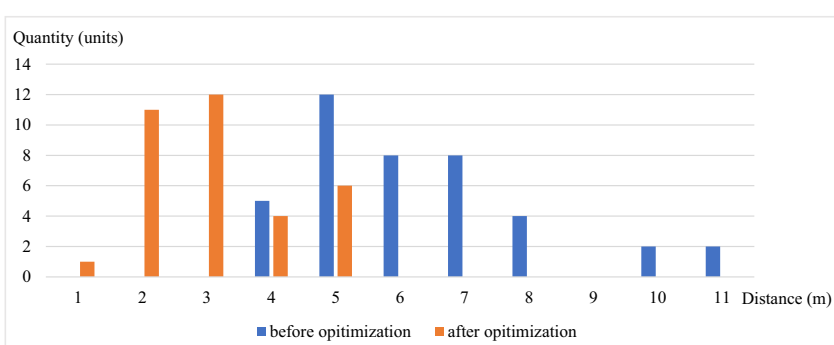
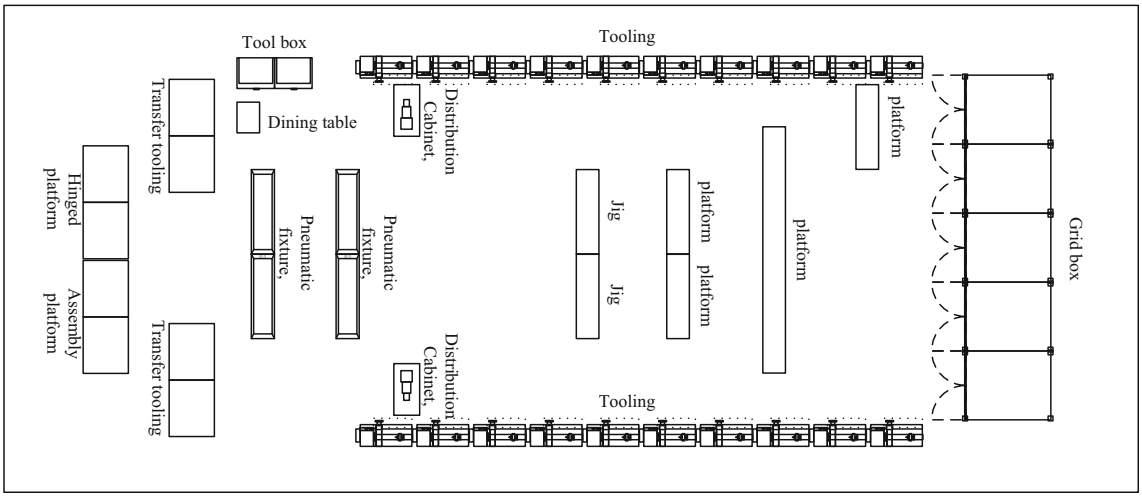
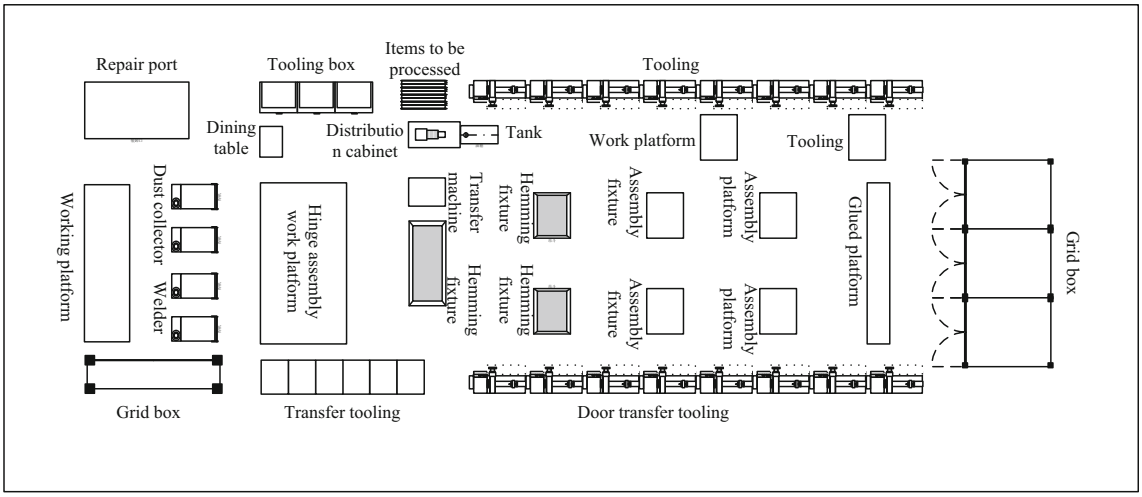


Fig. 7 The distance statistics between the stations of the door line



(a) Process layout plan of door line station before optimization



(b) Process layout plan of door line station after optimization

Fig. 8 Comparison of the process layout of the door line before and after optimization

5 Conclusions and future work

In this paper, we focus on a discrete manufacturing workshop layout optimization based on digital twin. First, a sub-framework of digital twin-based workshop partitioning is established. Entities in workshop are classified according to its characteristics to achieve the optimal partitioning of the workshop. Second, a sub-framework of digital twin-based equipment layout optimization is presented. The layout of equipment and facilities is adjusted through real-time data collection and value-added processing

Table 2 Optimization effect of digital twin-based discrete manufacturing workshop layout optimization method

Optimization object	Before optimization	After optimization	Optimization effect	Cost savings
Position appliances	-	-	Reduced 8 appliances	8800 yuan
Original material boxes	-	-	Save 60 original material boxes	4200 yuan
Work time	-	-	Reduced 114.5 min per day	11,846 yuan per year
Cycle time per piece	5 min and 35 s	5 min and 2 s	Reduced by 33 s	4038 yuan per year
Site layout and process flow	-	-	Save walking distance 67 steps per unit	3220 yuan per year
Floor, back, closing station	-	-	Save 15 min/day per shift	4588 yuan per year
Tooling	-	-	Reduced 51 sets of tooling	56,100 yuan
Work-in-process	82 sets	42 sets	Reduce work in progress by 40 sets	216,000 yuan

of twin data. Then, a sub-framework of digital twin-based distribution route optimization is developed. The tooling utilization is improved, and the WIP backlogs are eliminated by obtaining twin data and further making routing decision. Finally, a digital twin-based discrete manufacturing workshop layout optimization method is established based on the three sub-frameworks. Then, the proposed method is applied to a welding production workshop. The experimental results indicate that the production capacity is increased by 29.4%, the WIP is reduced by 51%, the amount of tooling is reduced by 41%, the working time is shortened by 114.5 min per day, and various expenses are saved by 350,000 yuan per year.

Thus far, there are few researches on the workshop layout optimization using digital twin. The layout optimization method in this paper can promote the workshop layout optimization in industrial practices. Future research on this topic needs to be undertaken on better combining digital twin technology with workshop layout optimization and standardizing and personalizing application methods.

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Data availability The data used to support the findings of this study are included within the article.

Compliance with ethical standards

Competing interests The authors declare that they have no competing interests.

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